

Quick Activity 2: Canteen Group Order

A canteen sells **rice meals at ₱85 each** and **drinks at ₱25 each**. Every order receives a **10% discount on the rice-meal subtotal only**. The canteen also charges a **₱5 packaging fee per rice meal**.

Develop an algorithm that calculates the final bill and the customer's change.

INPUT	PROCESSES	OUTPUTS
Number of rice meals Number of drinks Payment amount	Get the number of rice meals, drinks, and the payment. Find the rice-meal subtotal by multiplying the number of meals by ₱85.00. Find the drink subtotal by multiplying the number of drinks by ₱25.00. Calculate the 10% discount based only on the rice-meal subtotal. Subtract the discount from the meal subtotal. Calculate the packaging fee by multiplying the number of meals by ₱5.00. Add the discounted meal cost, drink subtotal, and packaging fee to get the final bill. Subtract the final bill from the payment to find the change. Display all monetary values with two decimal places.	Meal subtotal Drink subtotal Discount Packaging fee Final bill Change

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START

INPUT riceMeals

INPUT drinks

INPUT payment

mealSubtotal $\leftarrow$ riceMeals $\times$ 85
 drinkSubtotal $\leftarrow$ drinks $\times$ 25

discount $\leftarrow$ mealSubtotal $\times$ 0.10
 discountedMeals $\leftarrow$ mealSubtotal - discount

packagingFee $\leftarrow$ riceMeals $\times$ 5

finalBill $\leftarrow$ discountedMeals + drinkSubtotal + packagingFee
 change $\leftarrow$ payment - finalBill

DISPLAY mealSubtotal to 2 decimal places
 DISPLAY drinkSubtotal to 2 decimal places
 DISPLAY discount to 2 decimal places
 DISPLAY packagingFee to 2 decimal places
 DISPLAY finalBill to 2 decimal places
 DISPLAY change to 2 decimal places

END

DRY-RUN CALCULATION:QUICK ACTIVITY 2

Given: 3 rice meals
 4 drinks
 ₱500.00 payment

STEP	VARIABLE	CALCULATIONS	SOLUTIONS
1	RiceMeals	given	3
2	Drinks	given	4
3	payments	given	₱500.00
4	mealsubtotal	3 $\times$ ₱85.00	₱255.00
5	drinksubtotal	4 $\times$ ₱25.00	₱100.00
6	discount	₱255.00 $\times$ 0.10	₱25.50
7	packagingfee	3 $\times$ ₱5.00	₱15.00
8	finalbill	₱255.00 - ₱25.50 + ₱100.00 + ₱15.00	₱344.50
9	change	₱500.00 - ₱344.50	₱155.50

Final Answer – Test Case 1

Final Bill = ₱344.50

Change = ₱155.50

Test Case 2

Given: 5 rice meals

3 drinks

₱600.00 payment

STEP	VARIABLE	CALCULATION	SOLUTION
1	ricemeals	given	5
2	drinks	given	3
3	payments	given	₱600.00
4	mealsubtotal	$5 \times \text{₱}85.00$	₱425.00
5	drinkssubtotal	$3 \times \text{₱}25.00$	₱75.00
6	discount	$\text{₱}425.00 \times 0.10$	₱42.50
7	packagingfee	$5 \times \text{₱}5.00$	₱25.00
8	finalbill	$\text{₱}425.00 - \text{₱}42.50 + \text{₱}75.00 + \text{₱}25.00$	₱482.50
9	change	$\text{₱}600.00 - \text{₱}482.50$	₱117.50

Final Answer – Test Case 2

Final Bill = ₱482.50

Change = ₱117.50

EXPLANATION

Applying the 10% discount to the entire bill would give an incorrect result because the discount is only intended for the rice-meal subtotal. Drinks and packaging fees are not eligible for the discount.

For example, in Test Case 1, the correct discount is ₱25.50, which is 10% of the ₱255.00 meal subtotal. If the discount were applied to the entire bill before the discount, the

discount would be calculated on ₱370.00, resulting in ₱37.00. This would give the customer a discount on the drinks and packaging fee, which violates the rules.